

Isolation and purification of active constituents



Objectives

- List the different isolation techniques
- Explain the principle of classical methods
- Explain the principle of chromatography methods
- Describe the classification of chromatography

Natural product isolation

- How to isolate pure compound(s) (biological active material) from natural products.
- Chemicals known to have medicinal benefits are considered to be “active ingredients” or “active principles” of natural medicines.
- Natural products have provided the primary sources for new drug development.
- From the 1940s to the end of 2014, nearly half of the FDA approved chemical drugs for the treatment of human diseases were derived from or inspired by natural products

- Plant or animal tissues always contain several classes of compounds with markedly different structures.
- The utilization of whole plant or other crude preparations for therapeutic or experimental reasons can have several drawbacks including:
 - Co-occurrence of undesirable compounds
 - Variation in the amount of the active constituent with geographic areas, from one season to another
 - Changes or losses of bioactivity due to variability in collection, storage, and preparation of the raw material.

Extraction

- is the first step of phytochemistry research.
- The purpose of extraction is to get the objective chemical constituents to the utmost extent and avoid or reduce the solution of unwanted constituents
- A crude natural product extract is literally a cocktail of compounds.
- It is difficult to apply a single separation technique to isolate individual compounds from this crude mixture.

Fractionation

- A process of obtaining compounds of similar polarities of the division of a mixture into a number of discrete fractions.
- ✓ To ease the isolation process.

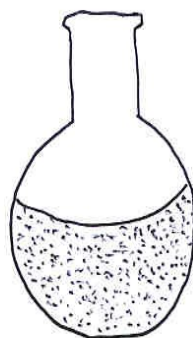
Plant



extraction



Crude Extract



Isolation

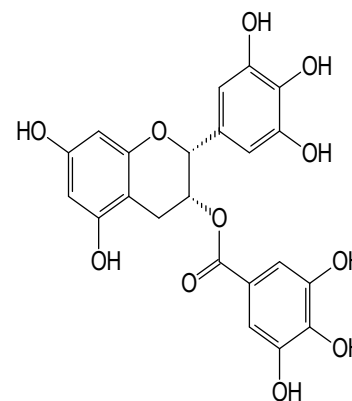


Pure Compound



=

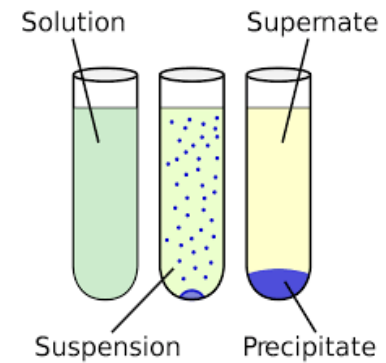
**Molecular
Structure**



- Isolation and purification methods can be:
 - Classical methods
 - Modern methods/Chromatography
 - The choice dépendes on...
 - ✓ Amount of sample
 - ✓ Level of purity required
 - ✓ Availability of facilities
 - ✓ Economical factors

A. Classical methods

1. Precipitation



- It is a method based on the formation of precipitation phytochemicals by reaction with specific reagents, or the precipitation of some components from the solution by adding specific reagents, which can reduce the solubility of some components in the solution.
- **Either compounds of interest or contaminants can be precipitated.**
- The precipitation reaction must be reversible if the target components are required to form precipitation.
- While if the components are nontarget, the precipitation generated will be removed, so the precipitation reaction can be irreversible.

- Substances may be precipitated from solutions either by:

decreasing solubility:

- by adding a solvent in which the compound of interest is not soluble.

e.g. Gums are soluble in water and they are usually separated from other water soluble contaminants by gradually adding methanol which leads to their precipitation

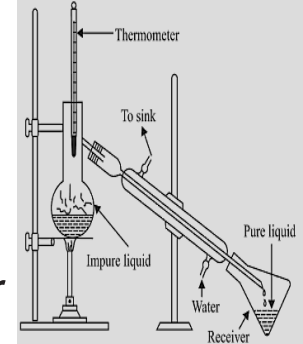
chemical reactions:

- tannins are may be precipitated from solutions by adding soluble salts of lead (e.g. lead acetate).



2. Simple distillation

- is a method of separating components in liquid mixtures based on their different boiling points.
- More volatile compounds are separated first and Less volatile compounds are separated at latter stages.
- becomes more and more difficult to separate compounds whose boiling points are very close.
- For example, the boiling points of the two alkaloids in total alkaloids of *Cicuta virosa*, coniine, and conhydrine are 166–167°C for the former and 226°C for the latter, which are quite different from each other, and then they can be separated by the fractional distillation method.

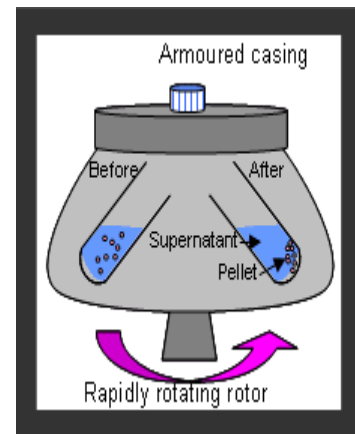


3. Formation of chemical derivatives

- Separation may be effected by making use of reversible reactions which can alter certain property of a substance.
- Alcohols or carboxylic acids (non-volatile) can be rendered volatile if converted to esters.
 - The esters can be hydrolyzed to regenerate the alcohols or carboxylic acids.
- When phenolic compounds which are organic soluble are treated with alkalis they form water-soluble phenolate salts.
 - Phenolate salts can generate back the original phenolic compounds if treated with acids.

4. Centrifugation

- Centrifugation is used for the separation of solid-liquid mixtures that are stubborn to settle or difficult to otherwise filter.
- More-dense components of the mixture migrate away from the axis of the centrifuge, while less-dense components of the mixture migrate towards the axis.
- This causes the solid to settle more rapidly and completely.



Chromatography

□ History

- Mikhail Tswett, Russian botanist invented the first chromatography technique in 1906.
- Used a column packed with a stationary phase of calcium carbonate to separate colored pigments from plant.
- He named this new technique chromatography based on the fact that it separated the components of a solution by color.
- *Chroma*-color, *graphy*-writing



- Is a physical method of separating the components of mixtures that are distributed between a stationary phase and mobile phase.
- The key to the separation is the differing affinities of analyte, among stationary phase and mobile phase.
- **Stationary phase:** the fixed or non-moving phase, immobile phase.
 - A solid or a liquid film on the surface of an inert support material.
- **Mobile phase:** is a moving phase that continuously flows through the stationary phase and carries the analyte.
 - Gas or liquid that carries the mixture of components through the stationary phase.

Classification

- **Mechanism of Separation**

1. Adsorption chromatography
2. Partition chromatography
3. Size exclusion chromatography/Gel filtration
4. Affinity chromatography
5. Ion exchange chromatography

- **Configuration of Instrument (supporter and carrier)**

1. Planar Chromatography (PC, TLC)
2. Columnar Chromatography (GC, HPLC)

- **Nature of the Mobile Phase**

1. Gas chromatography (GC)
2. Liquid chromatography (LC)

- **Polarity of the Stationary Phase**

1. Normal Phase
2. Reversed phase

- **Purpose of the Chromatographic Process**

1. Analytical Chromatography
2. Preparative Chromatography

Based on mechanism of action

A) Adsorption Chromatography

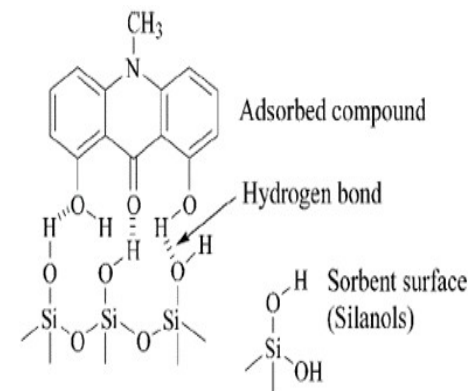
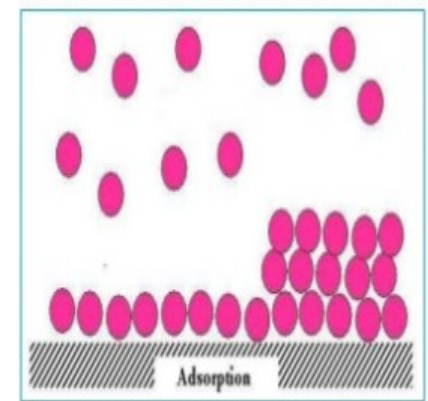
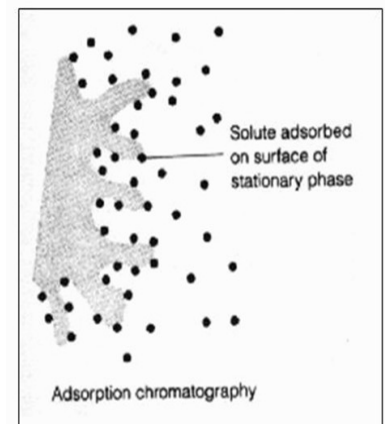
- It is a kind of chromatography based on the difference of adsorptive capacity of adsorbents to different compounds.

Adsorption versus Absorption:

- ☐ In absorption one substance penetrates into the bulk of another substance.
 - ✓ A sponge and water
- ☐ Adsorption is a surface phenomenon where interaction takes place only on the surface of one substance
 - ✓ The Packets of Silica Gel in New Shoes

- a type of chromatography where the solute molecules are directly bound to the surface of the stationary phase.
 - **SP**: solid (active sites on the surface of the solid stationary phase) and
 - **MP**: liquid or gas
- **Examples:** The commonly used adsorbents include silica gel, alumina, activated carbon, polyamide, and so on.

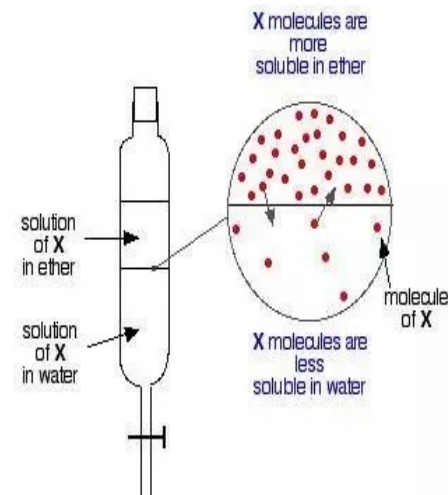
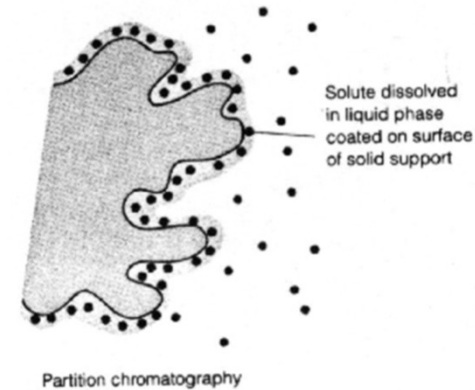
- Used mainly in **TLC & CC** packed with silica gel/alumina.
- Individual components move up at different rates, depending on intermolecular forces between:
 - the component and the stationary phase and
 - the component and the mobile phase.
- The active sites consist of
 - hydroxyl groups (for silica silanol groups, Si-OH)
- The interaction bond can be dipole-dipole, H-bond or London forces,...
- The affinity to them is greater polarity of the molecules, and the greater their ability to form hydrogen bonding.



- More polar_analytes (solutes with better polarity) interact more strongly with the stationary phase and move very slowly.
- More non-polar analytes (solutes with less polarity) interact less strongly with the polar silica gel and more strongly with the less polar mobile phase and move rapidly.

B) Partition chromatography

- It is a kind of chromatography method to separate components by using **different partition coefficients between stationary phase and mobile phase**, which are immiscible liquids.
 - **SP**: liquid bonded to solid surface
 - **MP**: liquid or gas
- Solute distribution equilibrates between the mobile phase and the stationary liquid.
- Partition is a phenomena of l-l, l-g and g-g.
- Partition chromatography could be divided into
 - normal phase chromatography and
 - reverse phase chromatography.



- The polarity of stationary phase is stronger than that of mobile phase in normal phase partition chromatography, which is mainly used to separate polar and moderately polar molecular compounds.
- Carriers commonly used in normal phase distribution chromatography include silica gel, cellulose powder, etc.
- Silica gel with water content of more than 12% can be used as a carrier for partition chromatography because of its loss of adsorption.
- In reverse phase partition chromatography, the polarity of mobile phase is stronger than that of stationary phase. The commonly used stationary phase is **octadecylsilylated silica** (ODS).

C) Ion Exchange Chromatography

- The principle of separation is by reversible exchange of ions between the target ions present in the sample solution (MP) to the ions present on ion exchangers.
 - **SP**: ion-exchange resin
 - A column is used that is filled with a charged stationary phase on a solid support, called an ion-exchange resin.
 - **MP**: liquid
- Ion exchange chromatography is suitable for the separation of ionic compounds, such as alkaloids, amino acids, organic acids, peptides, and flavonoids.

- Solute ions of the opposite charge in the mobile liquid phase are attracted to the resin by electrostatic forces.
- Separation is obtained since different substances have different degrees of interaction with the ion exchanger due to differences in their charges, charge densities and distribution of charge on their surfaces.
- In ion exchange chromatography one can choose whether to bind the substances of interest and allow the contaminants to pass through the column, or to bind the contaminants and allow the substance of interest to pass through.
- For ion exchange to be efficient the resin must have a higher affinity for the ion in solution compared to the ion in the resin.

IEX can be categorized into two classes

- the cation-exchange chromatography and
- the anion-exchange chromatography
- The ion exchangers basically contain charged groups covalently linked to the surface of an insoluble matrix (fixed ions).
- The charged groups of the matrix can be positively or negatively charged.

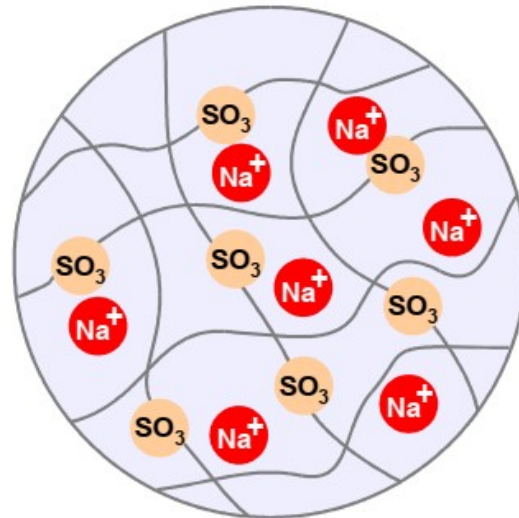
- For the cation-exchange
- In this process there are matrix (the resin), fixed charge and counter (active) ion
 - **Cationic exchangers** possess negatively charged group (fixed ions), and these will attract positively charged active ions (counter cation)
 - $S-X-C^+ + M^+ \leftrightarrow S-X-M^+ + C^+$
 - These exchangers are also called “Acidic ion exchange” materials, because their negative charges result from the ionization of acidic group like sulphonates (SO_3^-), carboxylate, phosphate, benzoate, and so forth

- the cation M^+ of the eluent replaced with the analyte cation C^+ bound to the anion X^- which is fixed on the surface of the chromatographic support (S).

□ They are usually (but not always) supplied in the Na^+ form:

□ $X-SO_3^-Na^+$

- The matrix= X^-
- Fixed charge= SO_3^-
- Counter ion=Na

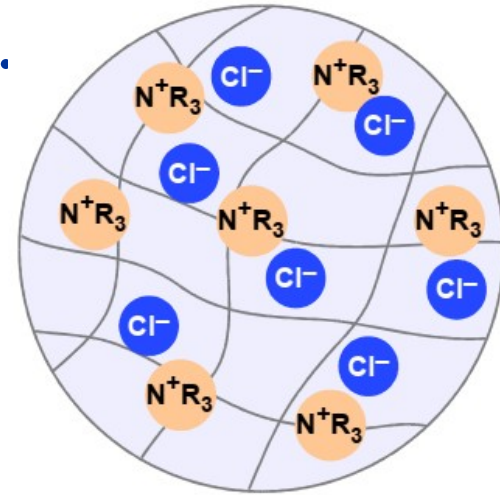


- For the **anion-exchange**
- **Anionic exchangers** have positively charged groups (fixed charge) that will attract negatively charged anions (counter (active) ions).
- $S-X^+A^- + B^- \leftrightarrow S-X^+B^- + A^-$
- The anion B^- of the eluent replaced with the analyte cation A^- bound to the positively charged ion X^+ on the surface of the stationary phase.
- These are also called “Basic ion exchange” materials like because their positive charges result from the ionization of basic group examples amino group, quaternary ammonium cations (NR_3^+), alkyl substituted sulphides (SR_2^+), ...

- The active counter ions can be supplied **as chloride.**
- Represented as: $\text{X-NR}_3^+\text{Cl}^-$
- The matrix= X-
- Fixed charge= $\text{NR}_3^+/\text{CH}_2\text{-N}^+\text{-(CH}_3)_3$
- Counter ion= Cl^-
- This is most effective method for water purification.

Common applications:

- Water treatment
- Ion recovery and removal from waste
- Separation of amino acids



D) MOLECULAR EXCLUSION CHROMATOGRAPHY

- Molecular sieve is the main principle of gel chromatography, which can separate mixture compounds according to the pore size of the gel and the molecular size of the compounds.
- Gel is a kind of solid material with a porous network structure.
- The molecules of the separated substances are different in size, so their ability to enter the gel is different.

- No attractive interaction between the solutes and the stationary phase:

Small molecules take longer time to pass through the column as they enter into the pores. The liquid passes through a porous gel (with no attractive action on solute molecules) which separates the molecules according to its **size**.

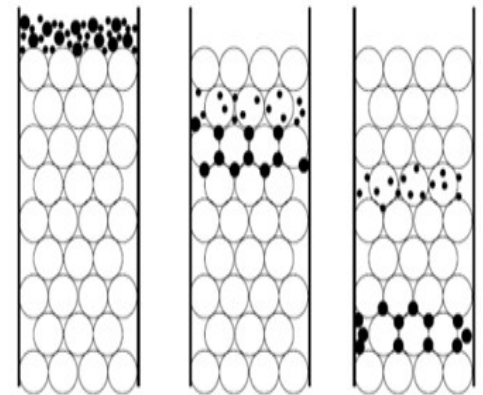
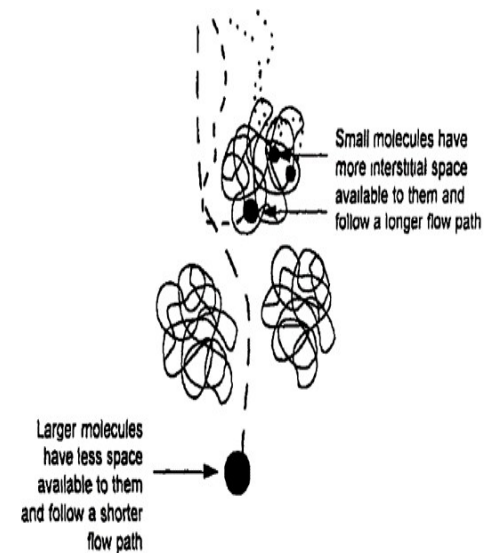
:Molecules separate according to their size

1. Smaller molecules enter the pores of the gel,.

2. Larger molecules pass through the column at a faster rate.

- There are many kinds of commercial gels, dextran gel and hydroxypropyl dextran gel are used most commonly

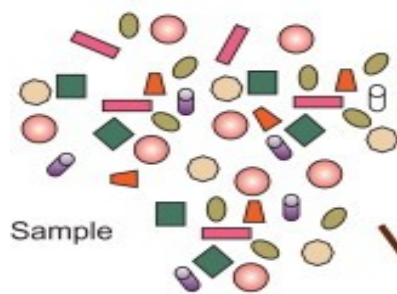
- When the mixture solution passes through the gel column:
- the molecules smaller than the gel pores can enter the gel interior freely, take longer time to pass through the column as they enter into the pores and need a larger volume of eluent,
- while the molecules with larger size than the gel pores cannot enter the gel, and only pass through the gel particle gaps.



E) AFFINITY CHROMATOGRAPHY

- It is a unique chromatographic separation method based on the principle of reversible combination of high affinity and specificity between molecules.
- Uses the adsorption medium coupled with affinity ligands as the stationary phase to adsorb target compounds while the target molecule is in the mobile phase.
- This method can selectively separate and analyze specific chemical constituents from complex samples.

- Firstly, ligands that can specifically bind to the target compounds are fixed on the filler carrier to make the chromatographic column.
- Then the mixture containing the target compounds is passed through the column.
- Only the target compounds which show affinity with the ligands can bind to the ligands and remain in the column.
- Finally, the adsorbed target compounds are eluted by changing the pH or ionic strength mobile phase.
- AC is mainly used for the separation and purification of proteins, especially enzymes, antigens, and antibodies



Sample

1



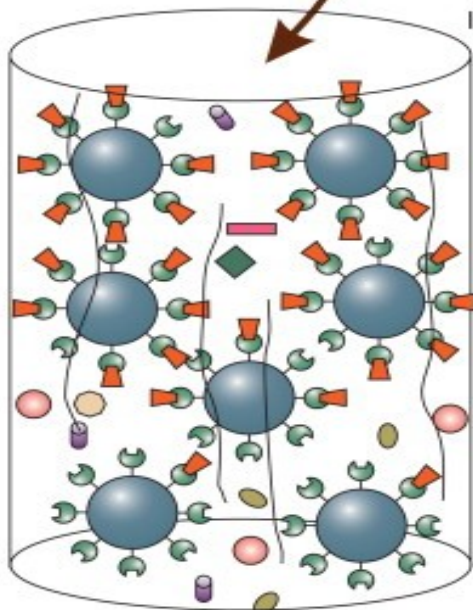
Affinity ligands are covalently attached to a chromatography support and packed in a column

2

The sample is flowed through affinity column in binding buffer

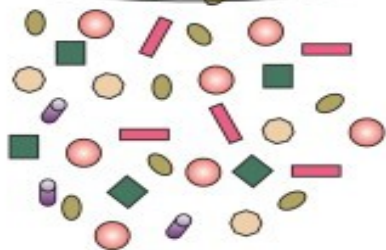
3

Target molecules bind to affinity ligands



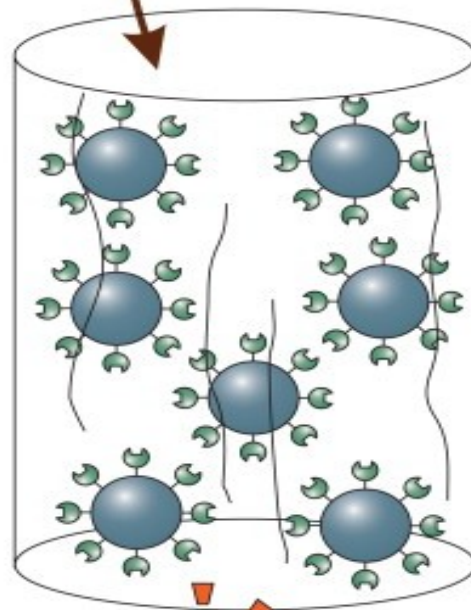
4

Sample is washed through the column to remove unbound components



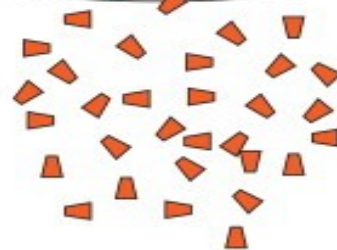
5

Elution buffer is applied to disrupt the interaction between the affinity ligand and the target molecule



6

Purified target molecule is eluted from the column



Based on Configuration of Instrument

A. Planar Chromatography

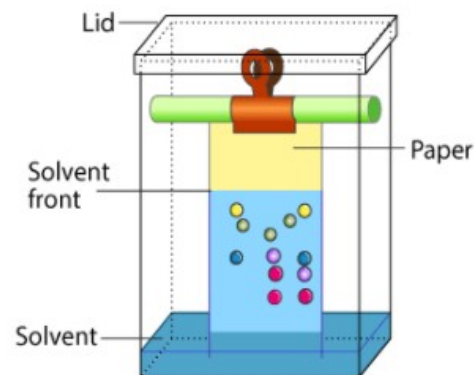
- Planar chromatography is one type of chromatography technique.
- the stationary phase is on a flat plate/plane and
- the mobile phase moves through stationary phase due to capillary action.
- There are two types of planar chromatography:
 - Paper Chromatography
 - Thin layer chromatography (TLC)

- Another **difference between planar and column chromatography** is that **planar chromatography** separates based on their **different** distances of travel **in a** set amount of time. **Column chromatography** separates solutes by their **different** times of travel along a given distance, the length of the **column**.

1. Paper Chromatography

- ☐ type of planar chromatography in which paper is used as stationary phase.
- ☐ Paper is comprised of cellulose, which is a polymer of the simple sugar glucose, and as such is very polar due to the –OH groups present in glucose.
- Because of the many exposed –OH groups, cellulose interacts strongly with polar water molecules.
- This interaction is so strong that “dry” paper is approximate 22% water by weight.
- It is this water adsorbed on the paper surface that is the stationary phase in paper chromatography.

- ❑ It is an example of a **partition chromatography**
 - ✓ **Liquid-Liquid chromatography**
- ❑ The first phase is the water, which is held in the pores of the cellulose filter paper used while
- ❑ The second phase is the mobile phase that moves over the paper holding the analyte.
- ❑ The factor governing separation of mixtures of solutes on cellulose paper is the **partition between two immiscible phases**.
- The compounds in the mixture get separated due to differences in their affinity towards water (in stationary phase) and mobile phase solvents during the movement of mobile phase under the capillary action of pores in the paper.



2. Thin Layer Chromatography

➤ A form of planar and

➤ Solid-liquid chromatography that involves:

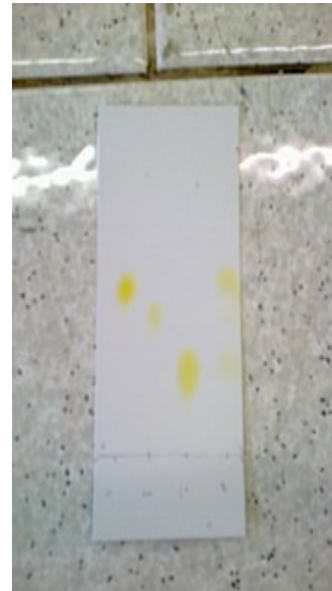
✓ **Stationary phase:** a thin layer of adsorbent material usually silica gel

✓ **Mobile phase:** liquid

- Separation is based on how the compounds interact with the adsorbent/SP.

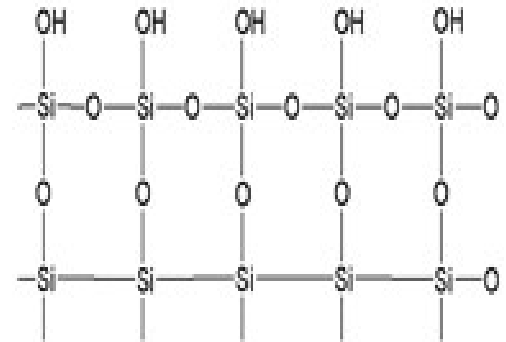
Types

- Analytical TLC
 - identify a substance
- Preparative TLC
 - Separate and isolate components in a mixture



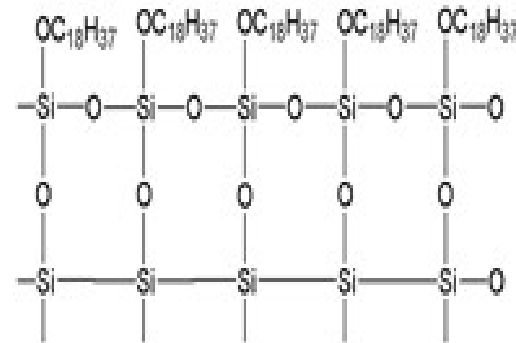
Normal phase TLC

- when the stationary phase is polar.
- the most polar compound has a stronger interaction with the silica and will not move higher.
- the less polar compound moves higher up the plate with the mobile phase.



Reversed phase

- When the stationary phase is silica bonded with an organic substrate such as long chain aliphatic acid like C-18.
- results from the adsorption of hydrophobic molecules onto a hydrophobic solid support in a polar mobile phase



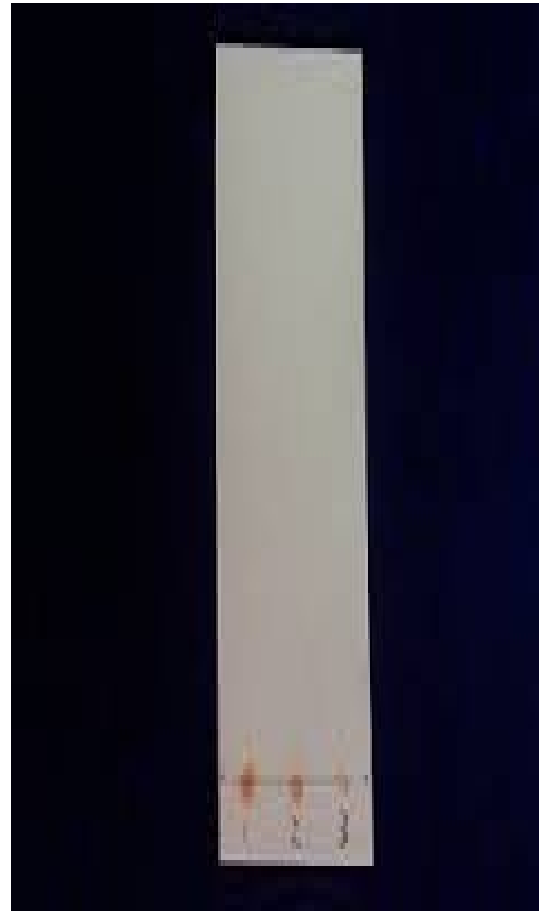
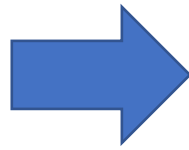
Steps for Running a PTLC

1. Plate preparation:

- Adsorbent (silica gel) + Water = thick slurry
- Spread, pouring, dipping and spraying done on inert carrier sheet like glass:
 - analytical TLC 0.1–0.25 mm
 - preparative TLC around 0.25–2 mm
- 30 minutes at 110 °C in oven. Why???

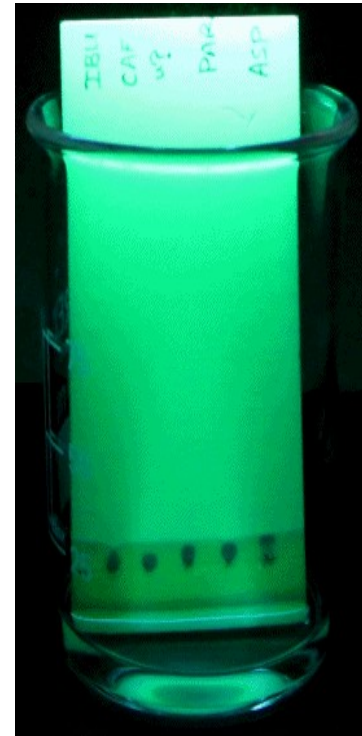
2. Sample Application

- A solution containing the sample is applied to a plate by using capillary tube,
 - About 1-1.5 cm from the base on 20 x 20 cm plate. Why ?
 - Application could be:
 - in form of spots (for analytical chromatography) or
 - in form of bands (for preparative work).
- Spots should not be closer than 1 cm center to center,
- 2-5 mm in diameter and 1-5 $\mu\text{g}/\mu\text{l}$.



3. Development

- The atmosphere in the tank has to be **saturated** with the solvent vapor by:
 - Filter paper lined the wall
 - keeping the tank for 30 minutes before putting the plate
- The plate is dipped in to a suitable solvent or solvent system and placed in a sealed container.

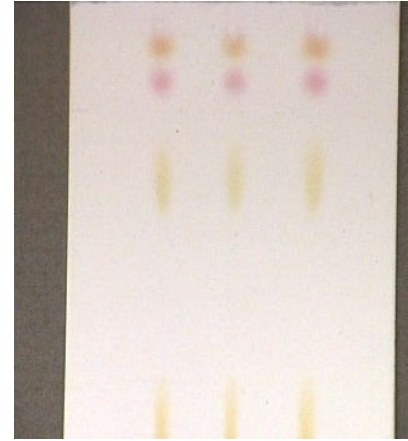


- The solvent moves up the plate by capillary action, dissolving the sample and moves up the plate.
- If the SP is polar and MP is non-polar
 - ✓ The most polar compound will have stronger interaction with the silica and
 - ✓ The less polar compound moves higher up the plate.
- For Non-polar stationary phase (reverse phase)
 - the mobile phase is changed to a more polar solvent
- Mixtures of water or aqueous buffers and organic solvents are used as mobile phase
 - ✓ The appropriate solvent in context of TLC will be one which differs from the SP material in polarity.
 - ✓ The sample spot will grow radially if the same polarity.

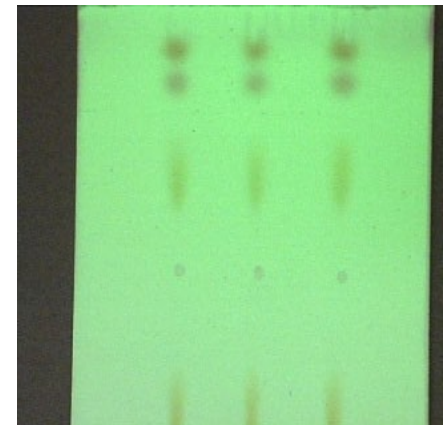
4. Analysis (Identifying the Spots)

A. Physical method of detection

1. If the spots can be seen (visible), outline them with a pencil.
 - ✓ Chromophore unit (Presence of conjugation)
2. If no spots are obvious, the most common visualization technique is to hold the plate under a UV lamp



Day light



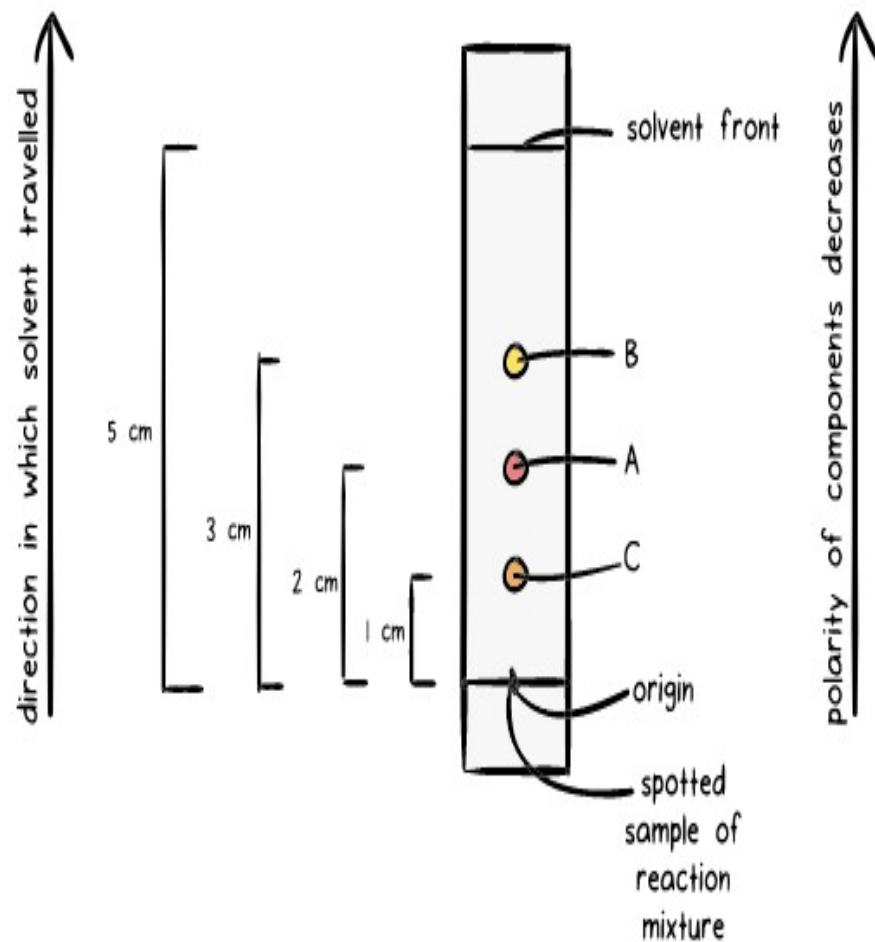
UV light

B. Chemical detection methods

- Spraying reagents/Chromogenic reagents.
 - General reagents: locate almost all of the existing organic compound classes e.g. H_2SO_4 , I_2
 - Specific reagents: only react with compounds containing a particular functional group e.g. Nihydriin, Dragendorff's

Retardation factor

- To quantify the results, the distance traveled by the substance being considered is divided by the total distance traveled by the mobile phase.
- Different compounds in the sample mixture travel at different rates.
- ✓ Differences in their attraction to the stationary phase



$$R_f = \frac{\text{Compound distance from origin}}{\text{Solvent front distance from origin}}$$

- Based on the R_f values, the identity and characteristics of the different compounds can be determined.
- **More polar compounds** will have **smaller R_f** values since they will have a stronger affinity for the polar solid phase.
- This would result in the compound not being carried very far along the TLC plate.
- Compounds with **larger R_f** values interact strongly with the less polar mobile phase and thus, tend to be **non polar** themselves.

Component	Distance travelled by the component (cm)	Distance travelled by the solvent (cm)	Retention factor (R_f) of the component
C	1	5	$R_{fC} = 1/5 = 0.2$
A	2	5	$R_{fA} = 2/5 = 0.4$
B	3	5	$R_{fB} = 3/5 = 0.6$

Based on the R_f values (as calculated above), component C is the most polar and component B is the least polar.

- R_f is a unique value for each compound under the same conditions.
- The R_f for a compound is a constant from one experiment to the next only if the chromatography conditions below are also constant:
 - solvent system
 - Adsorbent
 - thickness of the adsorbent
 - amount of material spotted
 - temperature

- The difference in R_f values between two spots on a plate, is used as a measure of the performance of the separation.
- The bigger the difference, the better the separation.
- For two closely migrating components, optimum resolutions are usually obtained when the
- R_f 's of both compounds are between 0.3 and 0.7.
- The difference in R_f (ΔR_f) value between two spots should be at least 0.1.

Uses

- **Identify** – by spot known substances – authentic - next to unknown substances on the same plate.)
 - identify compounds present in a given mixture
- **Analyze** – examine a mixture, its components,
 - determine the amount of the a mixture and/or the components present in the sample
- **Isolation** – separate components in order to isolate one of interest for further study
- **To check the purity** –a pure sample will show only one spot.
- Assess compound polarity
 - It is also important for determining the correct solvent system with which to run a column chromatography in.

- Require minimum sample preparation.
- Sensitivity is less than microgram/sample.
- Simultaneous analysis of multiple standards and samples.
- Greater speed and better resolution

